# EE-102: Electric Circuit Analysis

# Lecture no 2: *Basic Concepts of Circuits and Networks*

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# Organization of Lecture

- 2 Basic Concepts of Circuits and Networks
  - 2.1 Basic Elements of Circuits
  - 2.2 Node, Branches and Closed Path
  - 2.3 Voltage and Current Sources
  - 2.4 Ideal and Practical Sources
  - 2.5 Ohm's Law
  - 2.6 Kirchhoff's Voltage Law
  - 2.7 Kirchhoff's Current Law
  - 2.8 Solved Problems

# 2.1 Basic Elements of Circuits

## <u>Circuits and Networks</u>

- An electric circuit consists of Resistors **(R)**, Inductors **(L)**, Capacitors **(C)**, voltage sources and/or current sources connected in a particular combination. When the sources are removed from a circuit, it is called a **network**.

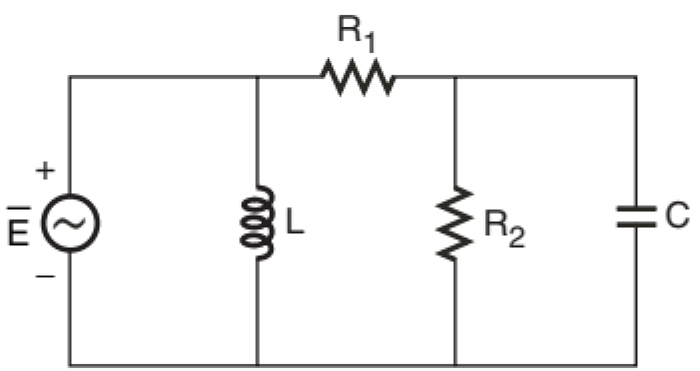



***Fig. a :*** *Circuit.*

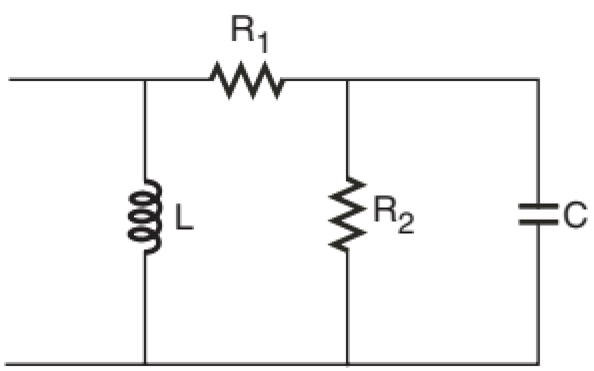



***Fig. b :*** *Network.*

***Figure 1: Example of circuit and network***

# 2.1 Basic Elements of Circuits (continued…)

## DC Circuits

- The networks excited by dc sources are called **dc circuits**. In a dc source, the voltage and current do not change with time. Hence, the property of capacitance and inductance will not arise in steady state analysis of dc circuits. Therefore, only resistive circuits are discussed in this lecture.

## Active and Passive Elements

- The elements of a circuit can be classified into active elements and passive elements. The elements which can deliver energy are called **active elements**. The elements which consume energy either by absorbing or storing are called **passive elements**.
- The active elements are voltage and current sources. The sources can be of different nature. The sources in which the current/voltage does not change with time are called **direct current sources** or in short **dc sources**. (But in dc sources, the current/voltage changes with load). The sources in which the current/voltage sinusoidally varies with time are called **sinusoidal sources** or **alternating current** sources or in short **ac sources.**

## 2.1 Basic Elements of Circuits (continued…)

- The passive elements of a circuit are resistors, inductors and capacitors, which exhibit the property of resistance, inductance and capacitance, respectively under ideal conditions. Resistance, inductance and capacitance are called **fundamental parameters** of a circuit. Practically, these parameters will be distributed in nature. For example, the resistance of a transmission line will exist throughout its length. But for circuit analysis, the parameters are considered as lumped.
- The resistor absorbs energy (and the absorbed energy is converted into heat). The inductor and the capacitor store energy. When the power supply in the circuit is switched ON, the inductor and the capacitor store energy, and when the supply is switched OFF, the stored energy leaks away in the leakage path. (Hence, inductors and capacitors cannot be used as storage devices).

# 2.1 Basic Elements of Circuits (continued…)


***Fig. a :*** *dc voltage source.*

***Fig. b :*** *dc current source.*

***Fig. c :*** *ac voltage source.*

***Fig. d :*** *ac current source.*

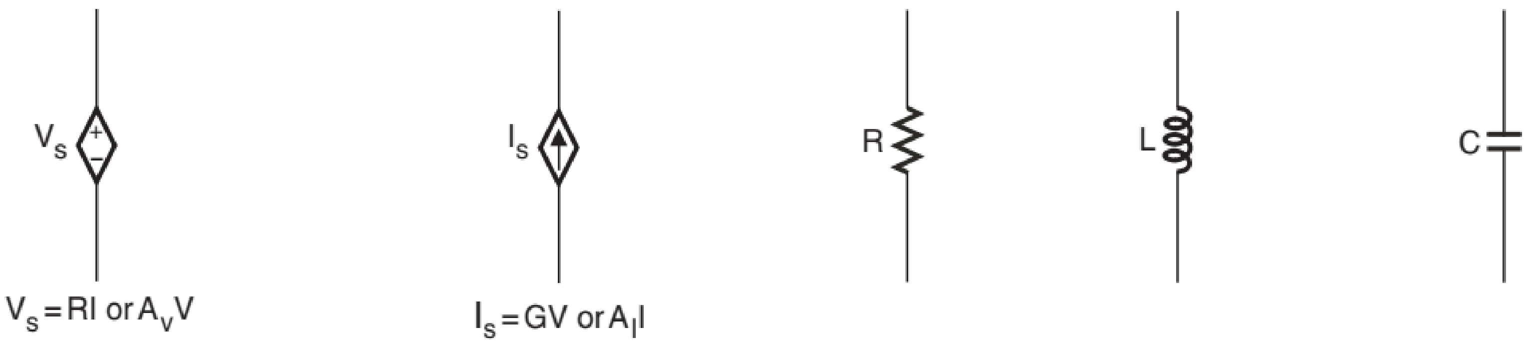



***Fig. e :*** *Dependent voltage source.*

***Fig. f :*** *Dependent current source.*

***Fig. g :*** *Resistance.*

***Fig. h :*** *Inductance.*

***Fig. i :*** *Capacitance.*

***Figure 2: Symbols of active and passive elements of circuits***

# 2.1 Basic Elements of Circuits (continued…)

**Independent and Dependent Sources**

- Sources can be classified into independent and dependent sources. The electrical energy supplied by an independent source does not depend on another electrical source. Independent sources convert energy in some form into electrical energy. For example, a generator converts mechanical energy into electrical energy, a battery converts chemical energy into electrical energy, a solar cell converts light energy into electrical energy, a thermocouple converts heat energy into electrical energy, etc.
- The electrical energy supplied by a dependent source depends on another source of electrical energy. For example, the output signal (energy) of a transistor or op-amp depends on the input signal (energy), where the input signal is another source of electrical energy.
- In the circuit sense, the voltage/current of an **independent source** does not depend on voltage/current in any part of the circuit. But the voltage/current of a **dependent source** depends on the voltage/current in some part of the same circuit.

# 2.2 Nodes, Branches and Closed Path

- A typical circuit consists of lumped parameters, such as resistance, inductance, capacitance and sources of electrical energy like voltage and current sources connected through resistance-less wires.
- In a circuit, the meeting point of two or more elements is called a **node**. If more than two elements meet at a node then it is called the **principal node**.
- The path between any two nodes is called a **branch**. A branch may have one or more elements connected in series.
- A **closed path** is a path which starts at a node and travels through some part of the circuit and arrives at the same node without crossing a node more than once.
- The nodes, branches and closed paths of a typical circuit are shown in Fig. 3 next slide(s). The nodes of the circuit are the meeting points of the elements denoted as A, B, C, D, E and F. The nodes A,B, C and D are principal nodes because these nodes are meeting points of more than two elements.

# 2.2 Nodes, Branches and Closed Path (continued…)

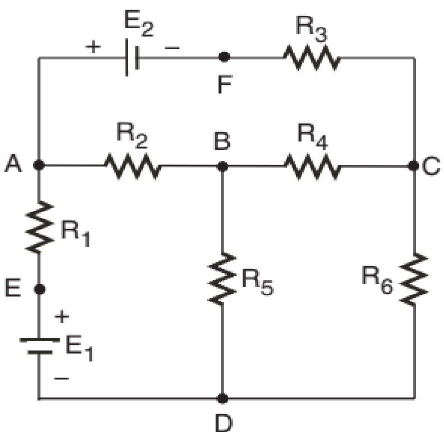



***Fig. a :*** *Typical circuit.*

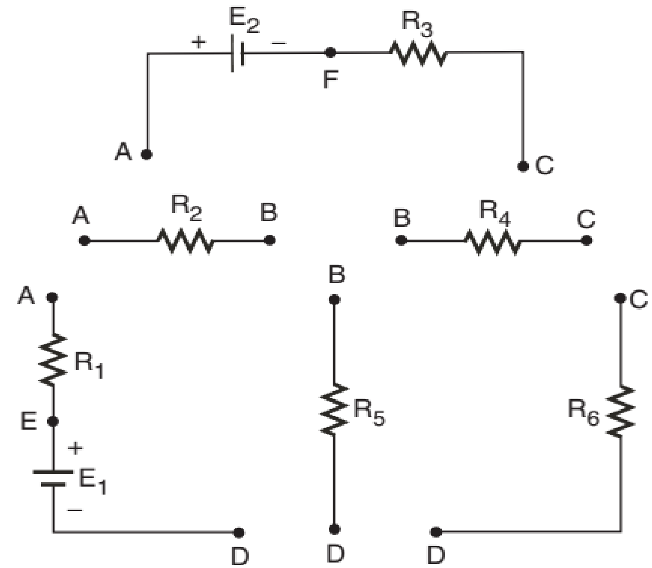



***Fig. b :*** *Branches of the circuit in Fig. a.*

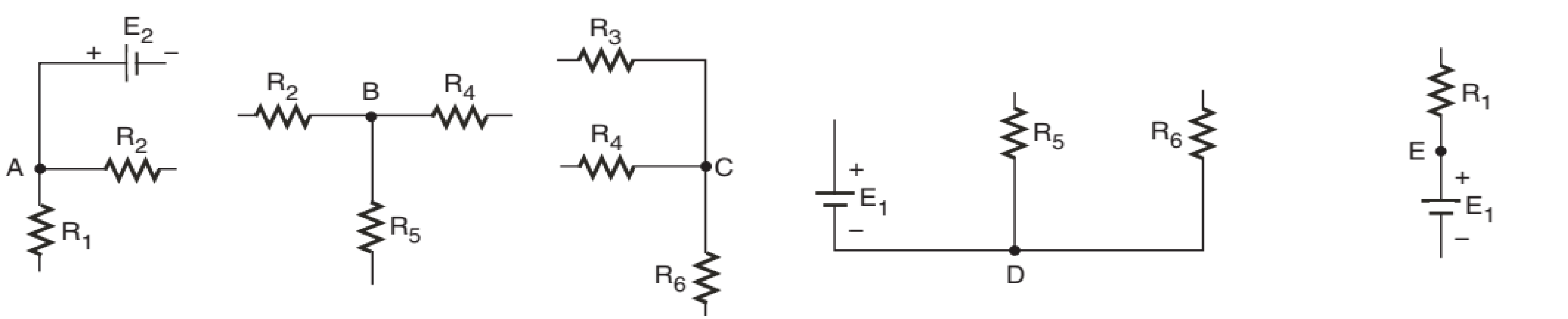



***Fig. c :*** *Nodes of the circuit in Fig. a.*


***Figure 3: A typical circuit and its branches, nodes and closed paths***

# 2.2 Nodes, Branches and Closed Path (continued...)

***Assume:***

- $R_4 >> R_2$
- $R_4 >> R_6$

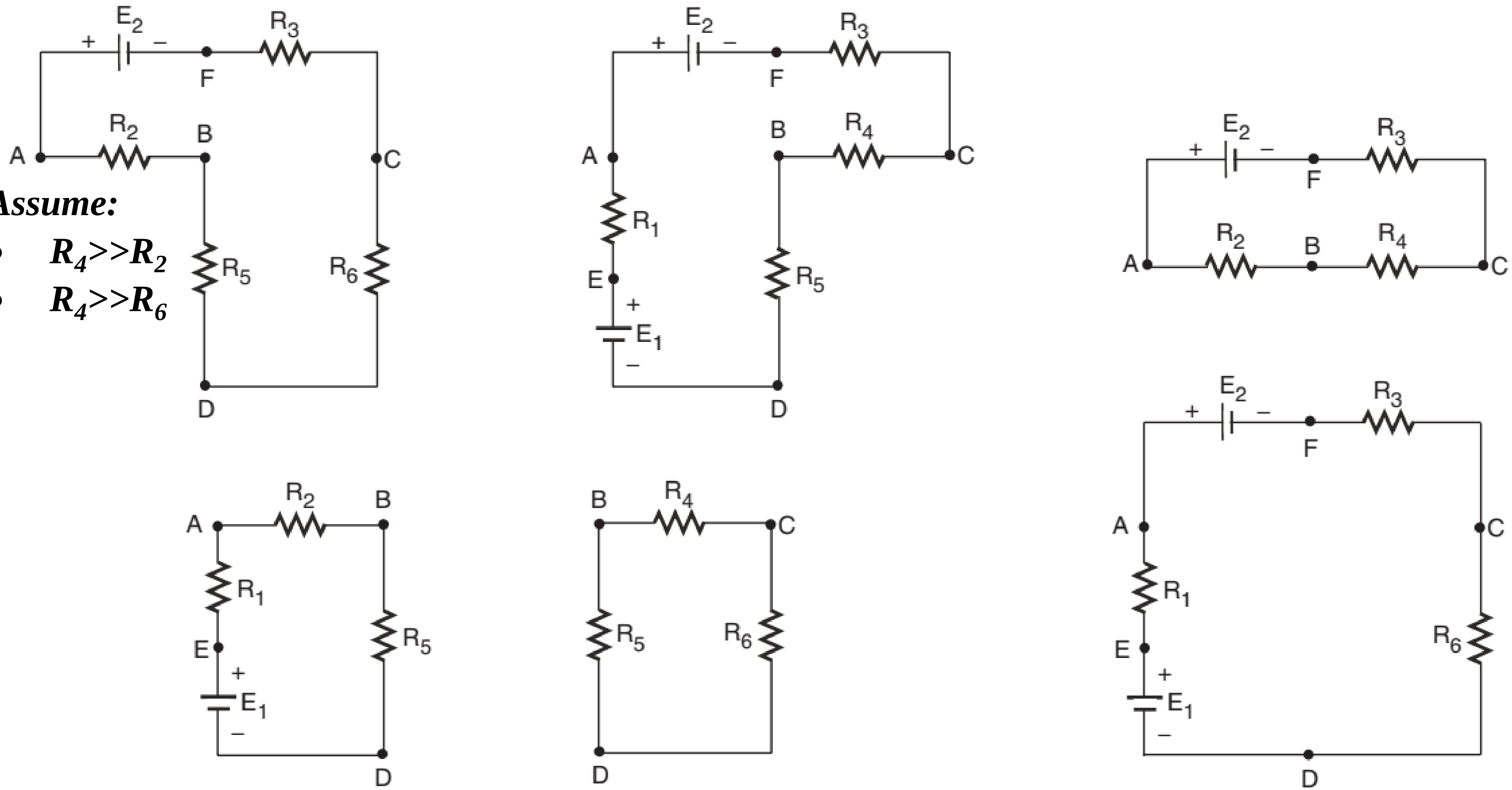



***Fig. d :*** *Closed paths of the circuit in Fig. a.*

***Figure 3: A typical circuit and its branches, nodes and closed paths (continued…)***

# 2.3 Voltage and Current Sources

- Voltage and current are two quantities that decide the energy supplied by the sources of electrical energy. Usually, the sources are operated by maintaining one of the two quantities as constant and by allowing the other quantity to vary depending on the load.
- When voltage is maintained constant and current is allowed to vary, the source is called a **voltage source**. When current is maintained constant and voltage is allowed to vary, the source is called a **current source**.

# 2.4 Ideal and Practical Sources (1/5)

- ***An ideal voltage source-*** It is a device that produces a constant voltage across its terminal, no matter what amount of current is drawn from it. i.e., terminal voltage($V_L$) is independent of load resistance($R_L$).

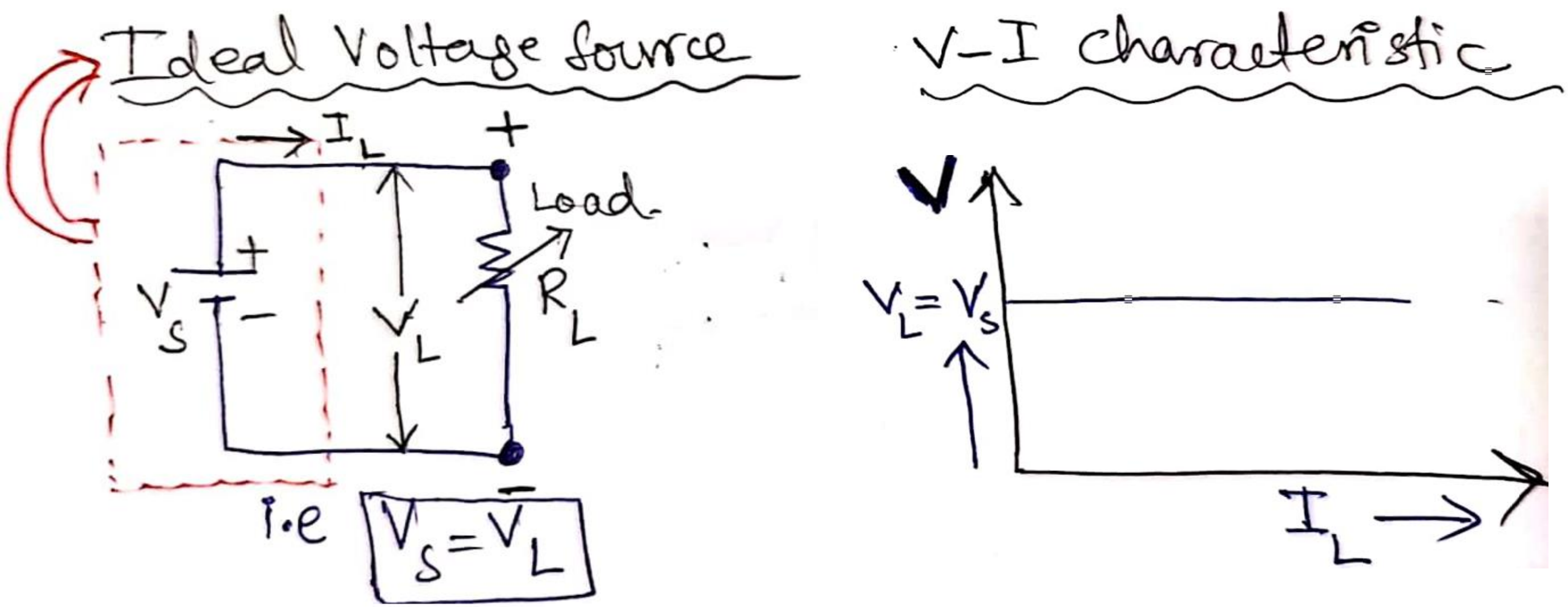
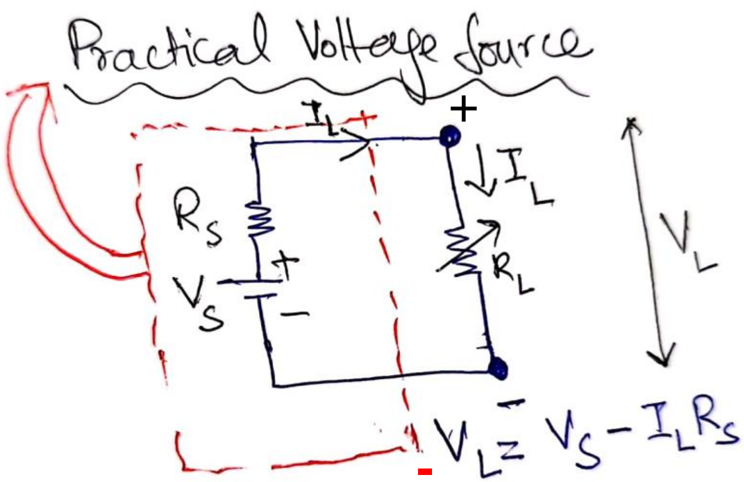



*Figure 5(a): Ideal Voltage Source*

# 2.4 Ideal and Practical Sources (2/5)

- In a ***practical voltage source***, the load resistance $\boldsymbol{R_L}$ connected across the source is decreased, the corresponding load current $\boldsymbol{I_L}$ increases, while the terminal voltage $\boldsymbol{V_L}$ across the source is increased).
- $\boldsymbol{V_L}$ is dependent of $\boldsymbol{R_L}$.
  - If $\boldsymbol{R_L}$ changes, $\boldsymbol{I_L}$ changes i.e., $\boldsymbol{I_L} = \boldsymbol{V_S} / (\boldsymbol{R_S} + \boldsymbol{R_L})$ .
    - When $\boldsymbol{I_L}$ changes, $\boldsymbol{V_L}$ changes i.e., $\boldsymbol{V_L} = \boldsymbol{V_S} - \boldsymbol{I_L}\,\boldsymbol{R_S}$




<u>V-I characteristic</u>

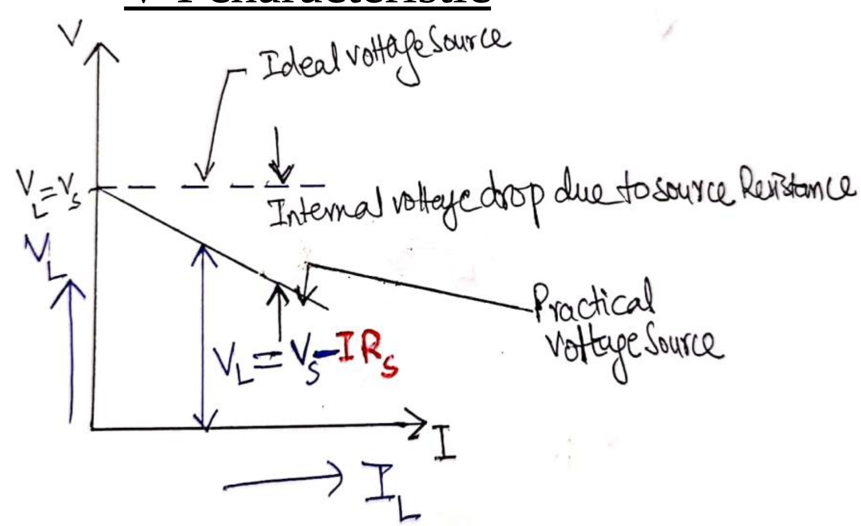



***Figure 5(b): Practical Voltage Source***

## 2.4 Ideal and Practical Sources (3/5)

- ***An ideal current source-*** is a device that delivers a constant current to any load resistance connected across it, no, no matter what the terminal voltage is developed across the load. . i.e., $\boldsymbol{I_S}$ is independent of $\boldsymbol{V_L}$ .

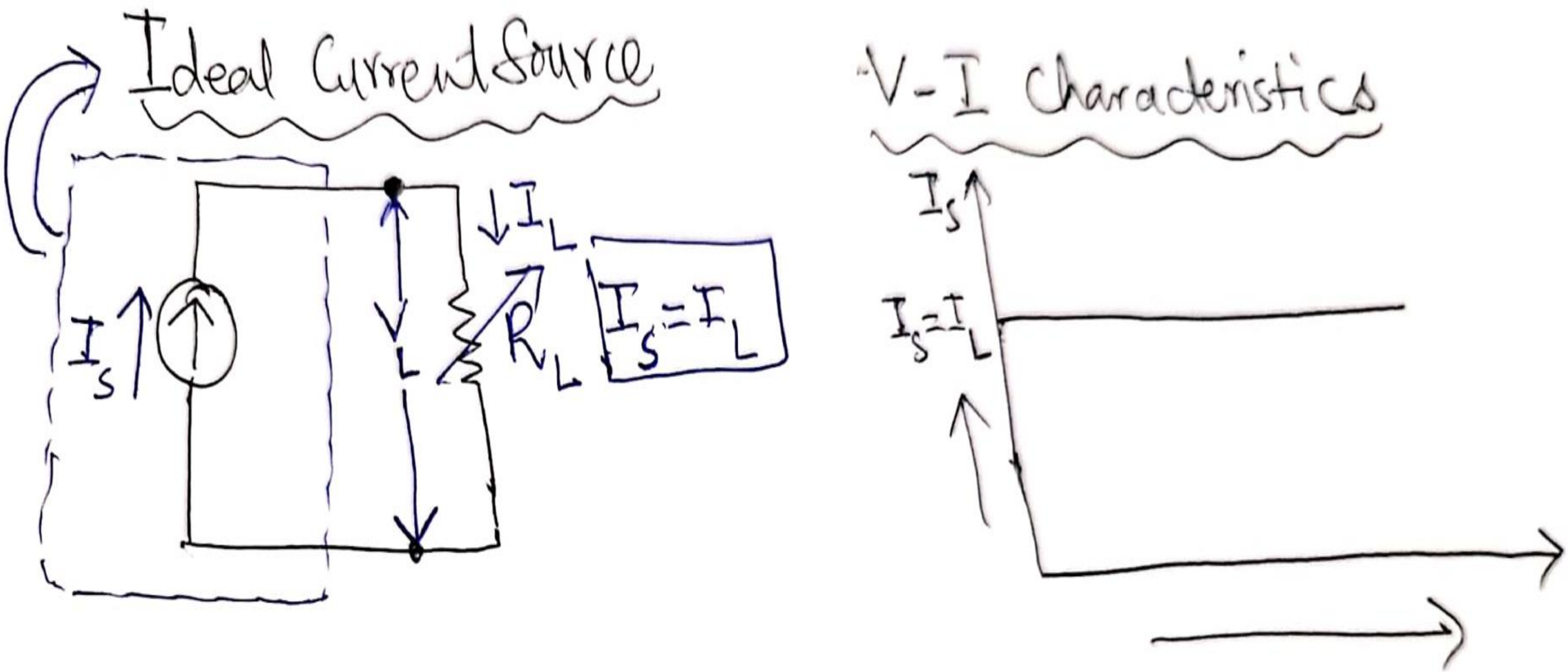
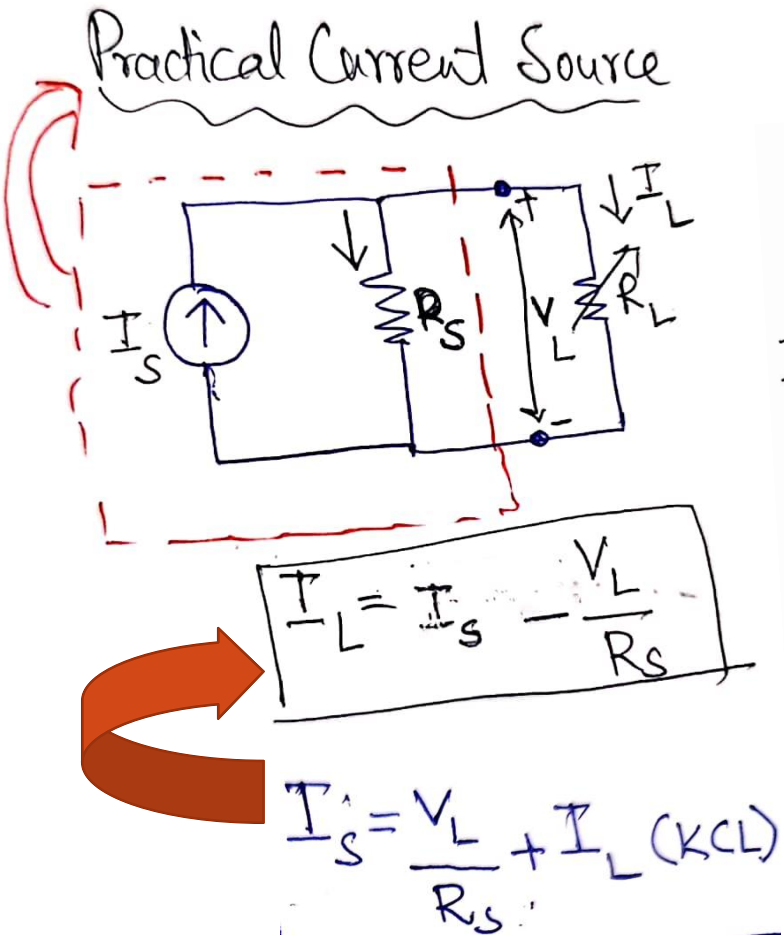



***Figure 5(c): Practical Voltage Source***

# 2.4 Ideal and Practical Sources (4/5)

- In a ***practical current source***, the current delivered by the source decreases with increasing load voltage and the reduction in current is due to its **internal resistance** (in parallel with current source).
- If $\boldsymbol{R_S}$ has a finite value, load current ($\boldsymbol{I_L}$) will never be constant, because some current will flow through $\boldsymbol{R_S}$ i.e., current is divided among $\boldsymbol{R_S}$ and $\boldsymbol{R_L}$.
- If $\boldsymbol{R_L}$ changes, $\boldsymbol{V_L}$ changes across $\boldsymbol{R_L}$.
- Practical current source is shown in figure 5(d) next slide.

# 2.4 Ideal and Practical Sources (5/5)




V-I characteristic

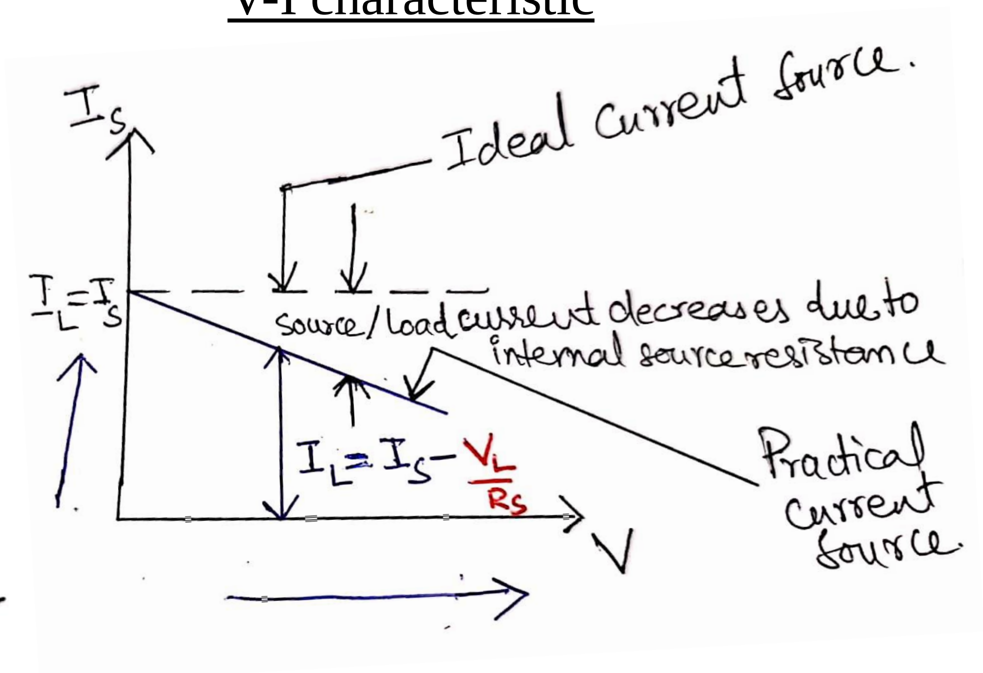



*Figure 5(d): Practical Current Source*

# 2.4 Ideal and Practical Sources (continued…)

- ***Practical Voltage Sources:*** Practical Voltage sources always have some series resistance ,although in some cases this resistance is so small in comparison with other circuit resistance that it may effectively be ignored when determining the operation of the circuit.
- ***Practical Current Sources:*** Similarly, a practical constant-current source will always have some shunt (or parallel) resistance. If this resistance is very large in comparison with the other circuit resistance, the internal resistance of the source may once again be ignored. An ideal current source has an infinite shunt resistance. Figure 6 shows equivalent voltage and current sources.

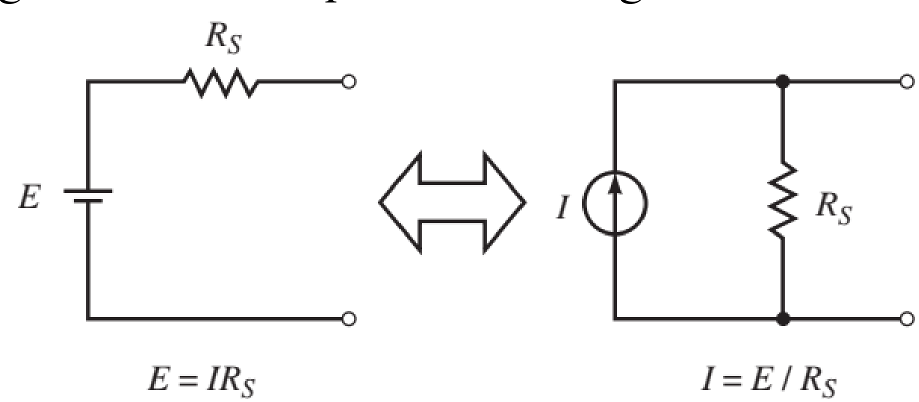



***Figure 6: Equivalent voltage and current sources***

# 2.4 Ideal and Practical Sources (continued...)

- If the internal resistance of a source is considered ,the source, whether it is a voltage source or a current source, is easily converted to the other type. The current source of Figure 6 (previous slide) is equivalent to the voltage source if:

$$I = \frac{E}{R_S}$$

and the resistance in both sources is $R_s$

- Similarly, a current source may be converted to an equivalent voltage source by letting:

$$E = IR_S$$

## 2.5 Ohm's Law

- *"The Current (**I**) passing through a conductor is directly proportional to the voltage (**V**) and inversely proportional to the Resistance (**R**)".*

$$\boxed{I = \frac{V}{R}}$$

**Where,**

**V - is Voltage (Volts)**

**I - is Current ( Amperes)**

**R - is Resistance**

# 2.5 Ohm's Law (continued...)

*All variations are called "**Ohm's Law**" and mathematically equal to one another.*

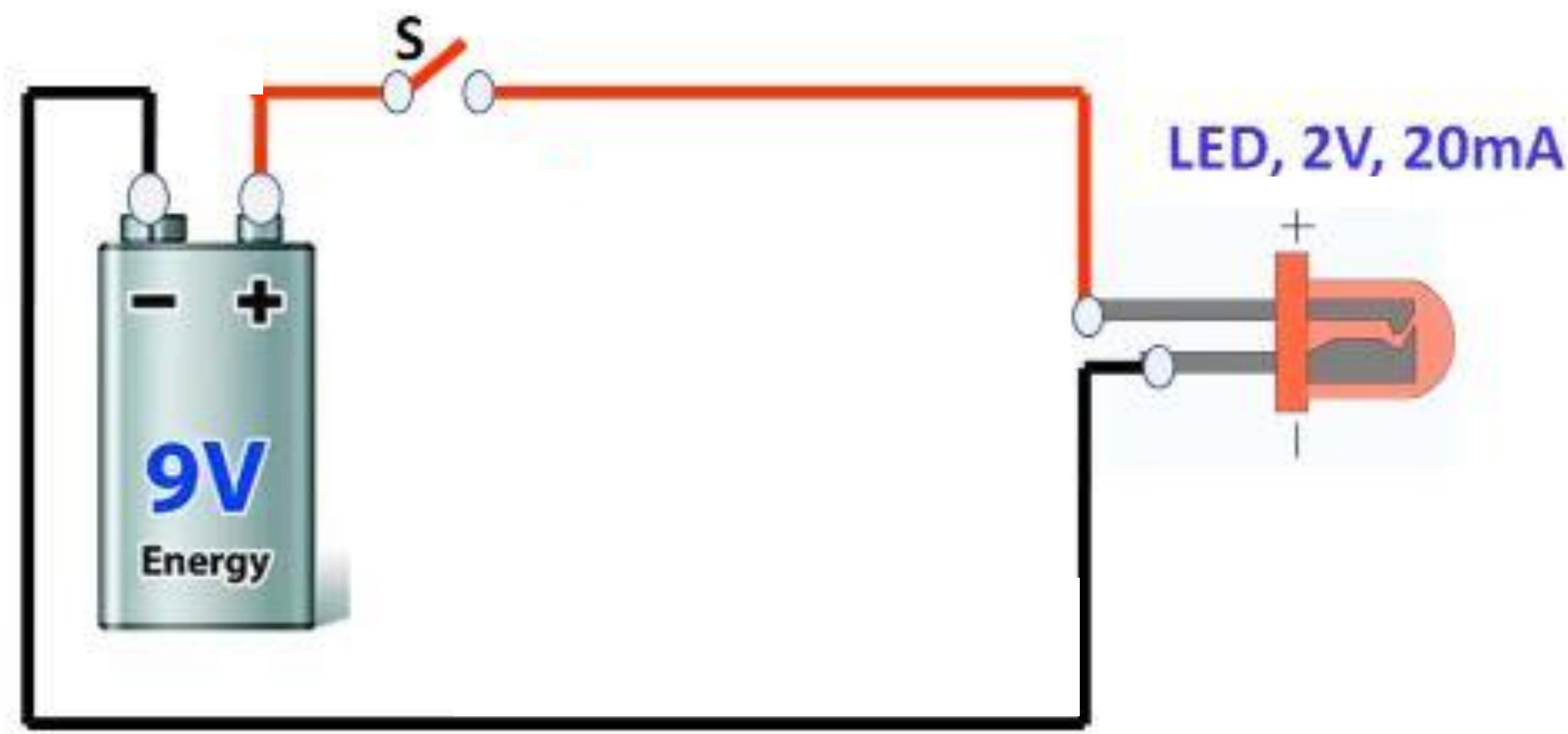
$$\boxed{V = I * R}$$

$$\boxed{I = \frac{V}{R}}$$

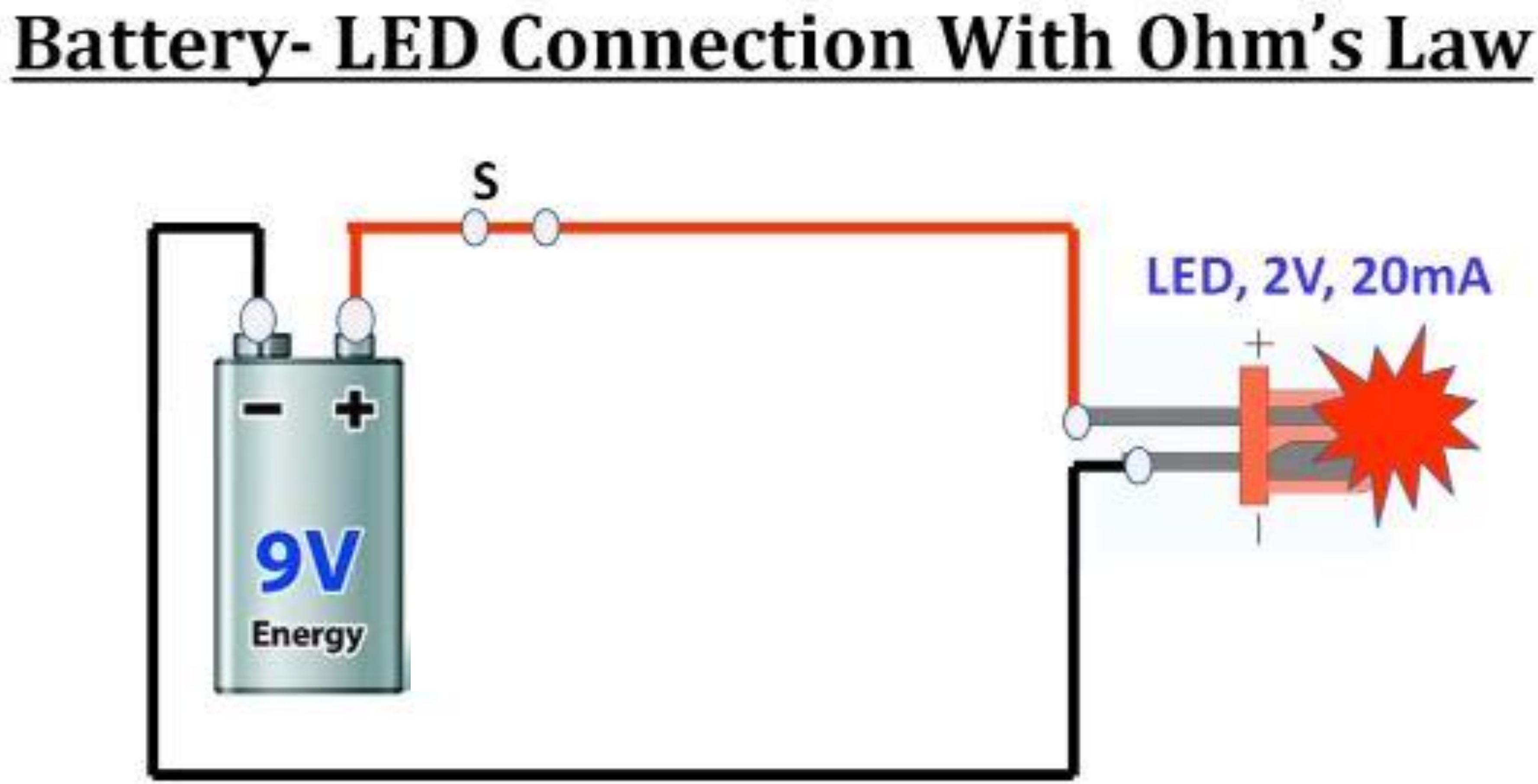
$$\boxed{R = \frac{V}{I}}$$

# 2.5 Ohm’s Law (continued…)

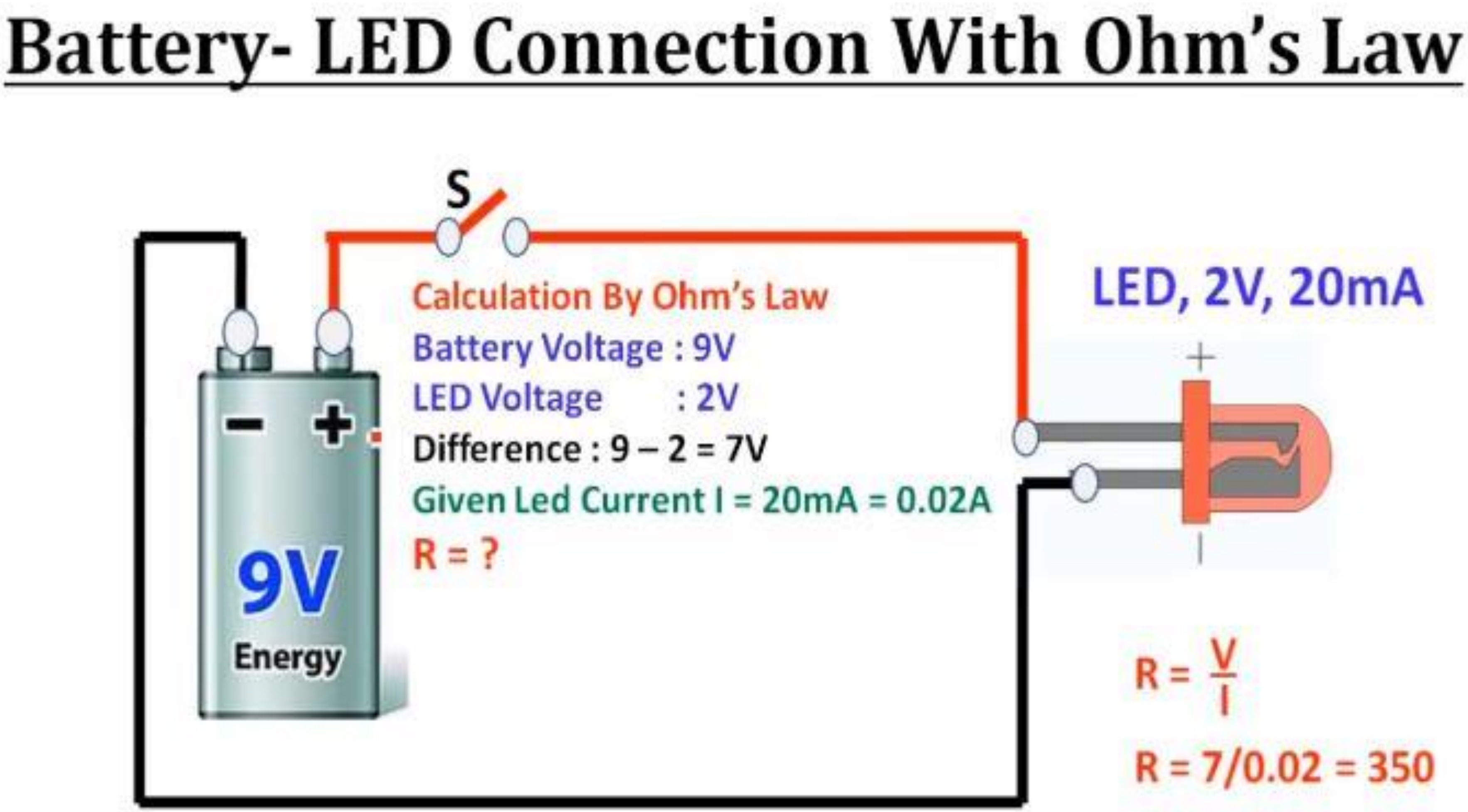
## Battery- LED Connection With Ohm’s Law



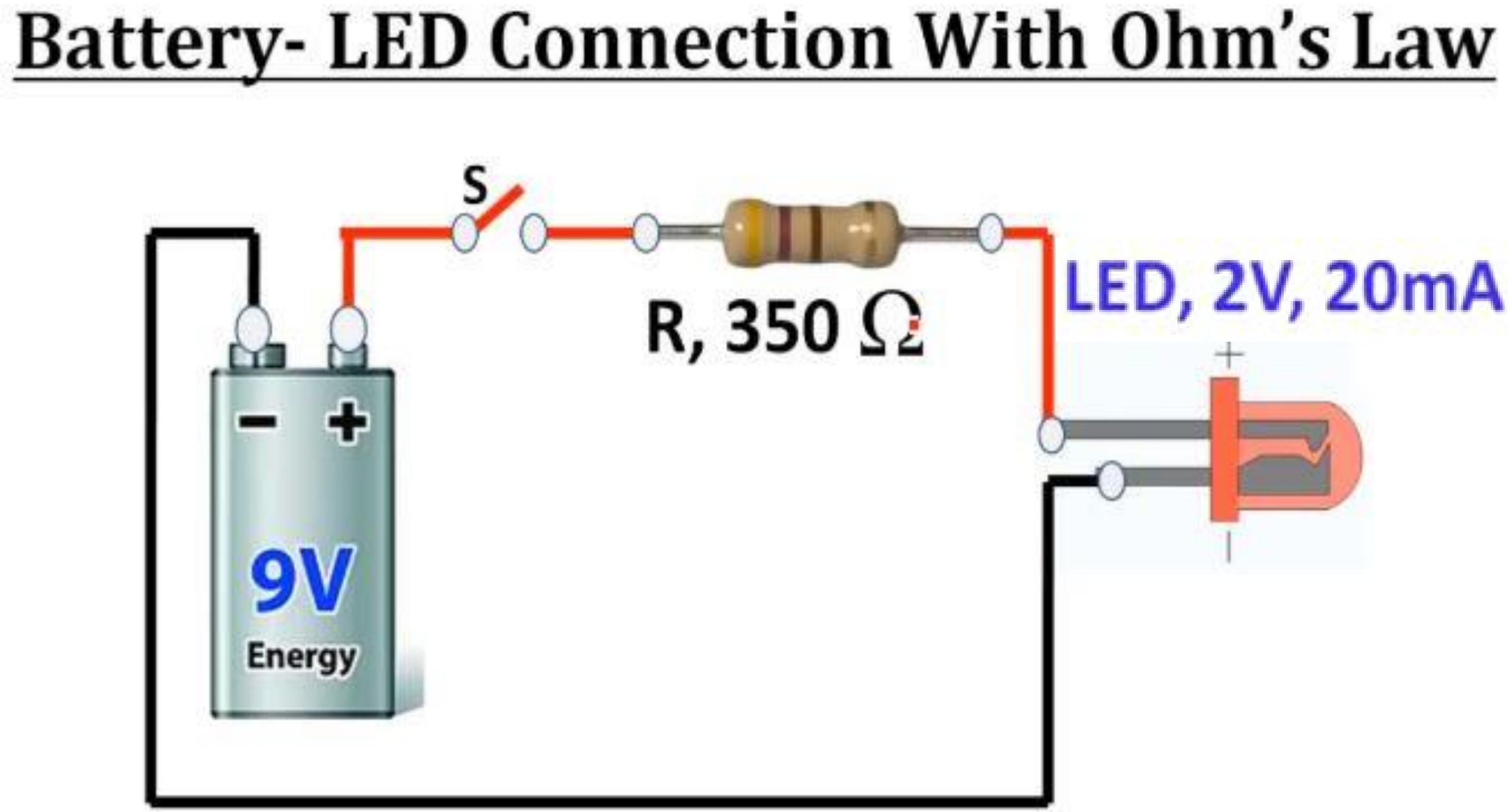
# 2.5 Ohm’s Law (continued…)



# 2.5 Ohm's Law (continued...)

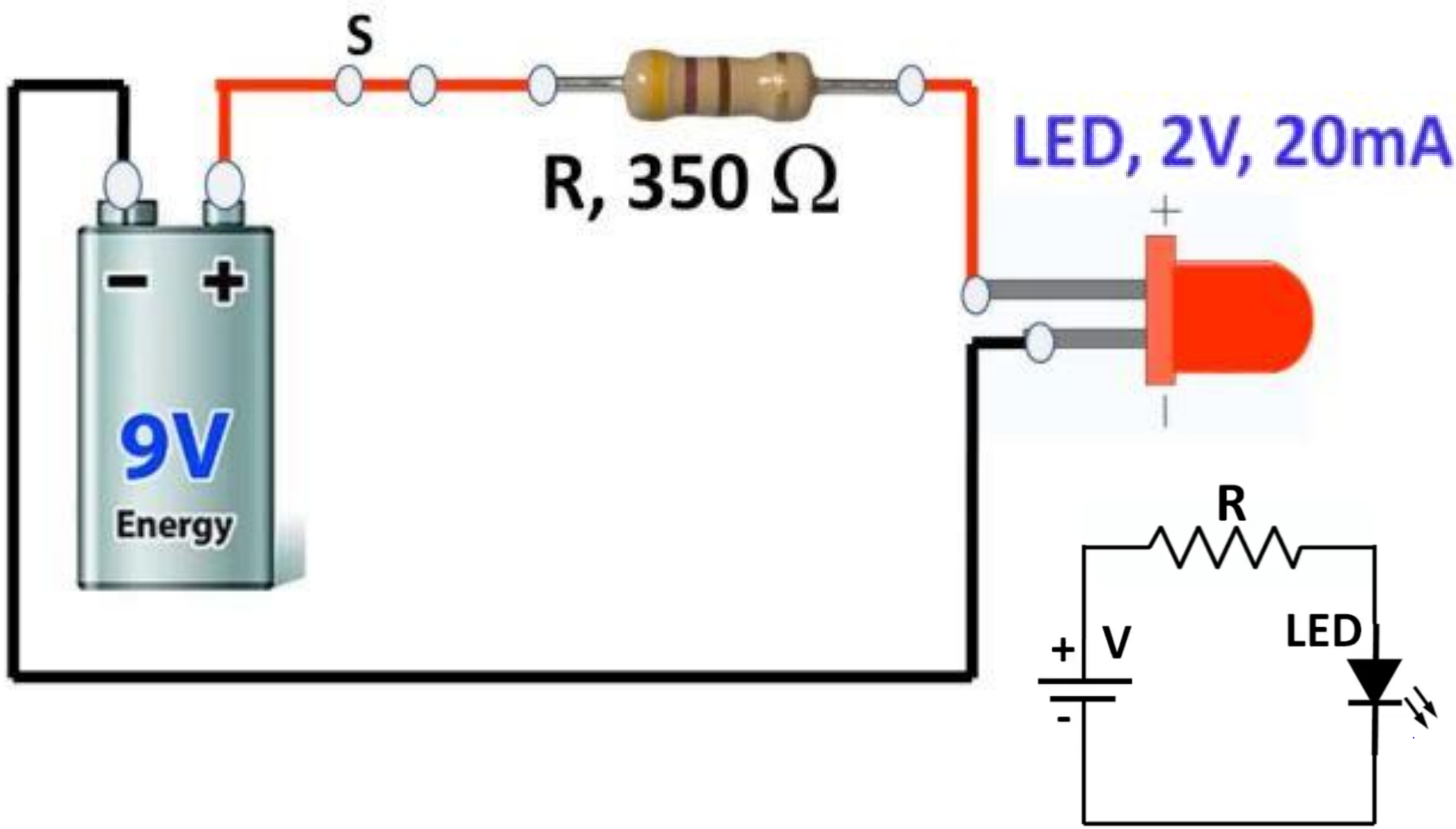


# 2.5 Ohm's Law (continued...)



# 2.5 Ohm's Law (continued...)

## Battery- LED Connection With Ohm's Law



# 2.6 Kirchhoff's Voltage Law

- Next to Ohm's law, one of the most important laws of electricity is **Kirchhoff's voltage law(KVL)**, which states the following:
  - *The summation of voltage rises and voltage drops around a closed loop is equal to zero. Symbolically, this may be stated as follows:*

$$\Sigma_{\circlearrowright} V = 0 \quad \text{for a closed loop}$$

- An alternate way of stating Kirchhoff's voltage law is as follows:
  - *The summation of voltage rises is equal to the summation of voltage drops around a closed loop.*

$$\Sigma_{\circlearrowright} V_{\text{rises}} = \Sigma_{\circlearrowright} V_{\text{drops}} \quad \text{for a closed loop}$$

# 2.6 Kirchhoff's Voltage Law (continued...)

- If we consider the circuit of Figure 7, we may begin at point *a* in the lower left-hand corner. By arbitrarily following the direction of the current, $I$ ,we move through the voltage source, which represents a rise in potential from point *a* to point *b*. Next, in moving from point *b* to point *c*, we pass through resistor $R_1$ which presents a potential drop of V1. Continuing through resistors $R_2$ and $R_3$, we have additional drops of $V_2$ and $V_3$ , respectively. By applying Kirchhoff's voltage law around the closed loop, we arrive at the following mathematical statement for the given circuit:

$$E - V_1 - V_2 - V_3 = 0$$

- Although we chose to follow the direction of current in writing Kirchhoff's voltage law equation, it would be just as correct to move around the circuit in the opposite direction. In this case the equation would appear as follows: $V_3 + V_2 + V_1 - E = 0$

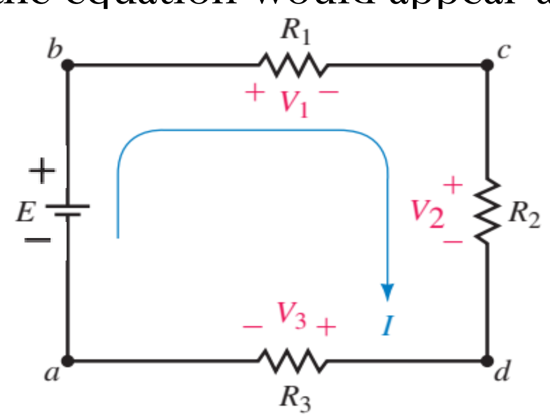



***Figure 7:** Kirchhoff's voltage law*

# 2.7 Kirchhoff's Current Law

- **Kirchhoff's Current law(KCL)**,which states the following:
  - *The summation of currents entering a node is equal to the summation of currents leaving the node.*
- In mathematical form Kirchhoff's current law is stated as follows:

$$\Sigma I_{\text{entering node}} = \Sigma I_{\text{leaving node}}$$

***Figure 8*** is an illustration of Kirchhoff's current law. Here we see that the node has two currents entering, $I_1 = 5$ A and $I_5 = 3$ A, and three currents leaving, $I_2 = 2$ A, $I_3 = 4$ A, and $I_4 = 2$ A.

$$\Sigma I_{\text{in}} = \Sigma I_{\text{out}}$$

$$5\text{ A} + 3\text{ A} = 2\text{ A} + 4\text{ A} + 8\text{ A}$$

$$8\text{ A} = 8\text{ A (checks!)}$$

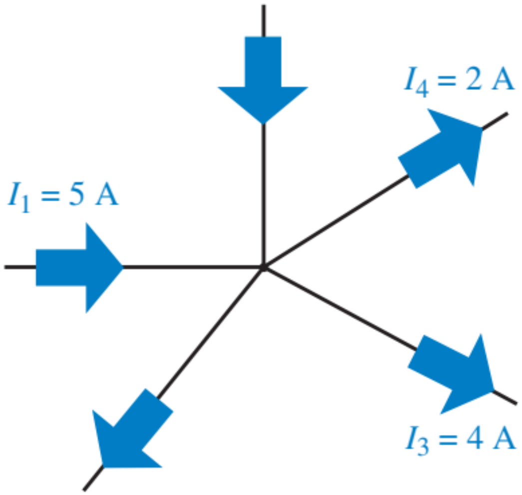



***Figure 8:*** *Kirchhoff's Current law*

## 2.8 Solved Examples

**Example 1** Use Kirchhoff's Voltage Law to determine the unknown voltage for the circuit in fig. 9:

**Solution:**

$$+E_1 - V_1 - V_2 - E_2 = 0$$

$$V_1 = E_1 - V_2 - E_2$$
$$= 16\text{ V} - 4.2\text{ V} - 9\text{ V}$$
$$V_1 = \mathbf{2.8\text{ V}}$$

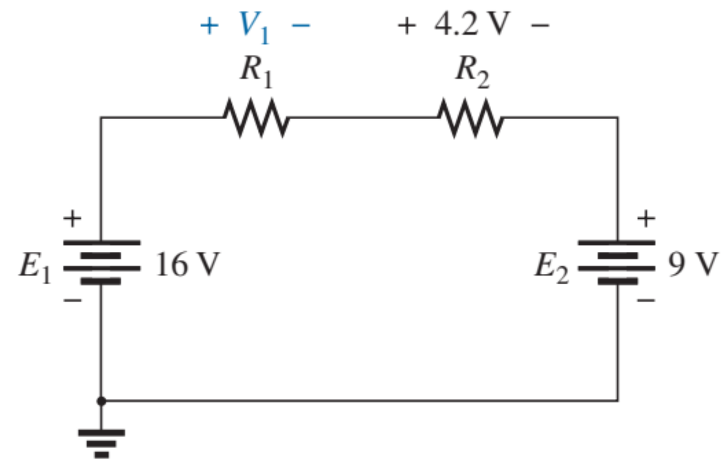



***Figure 9: Series circuit to be investigated in Example 1.***

The result clearly indicates that you do not need to know the values of the resistors or the current to determine the unknown voltage. Sufficient information was carried by the other voltage levels to determine the unknown.

## 2.8 Solved Examples

### *Homework!*

**Example 2** Using Kirchhoff's Voltage Law, determine the voltages $V_1$ and $V_2$ for the network in fig. 10:

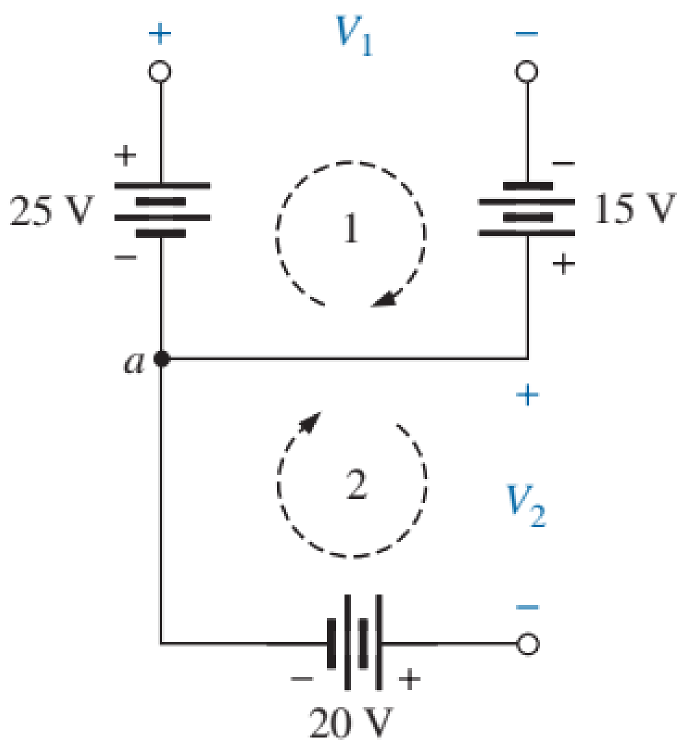



***Figure 10: Combination of voltage sources to be examined in Example 2.***

# 2.8 Solved Examples

**Example 3** Determine currents $I_3$ and $I_4$ in Fig 11 using Kirchhoff's Current Law:

## Solution:

At node $a$

$$\Sigma I_i = \Sigma I_o$$
$$I_1 + I_2 = I_3$$
$$2\text{ A} + 3\text{ A} = I_3 = \mathbf{5\text{ A}}$$

At node $b$, using the result just obtained,

$$\Sigma I_i = \Sigma I_o$$
$$I_3 + I_5 = I_4$$
$$5\text{ A} + 1\text{ A} = I_4 = \mathbf{6\text{ A}}$$

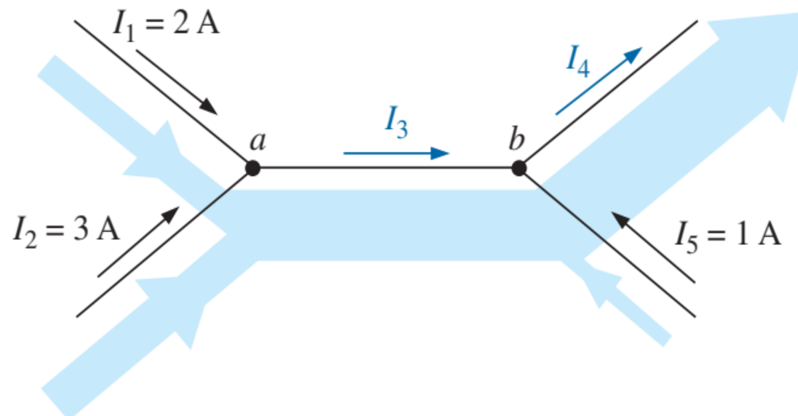



***Figure 11: Two-node configuration for Example 3.***

# 2.8 Solved Examples

***Homework!***

**Example 4** Determine currents $I_1$, $I_3$, $I_4$ and $I_5$ in fig. 12 using Kirchhoff's Current Law:

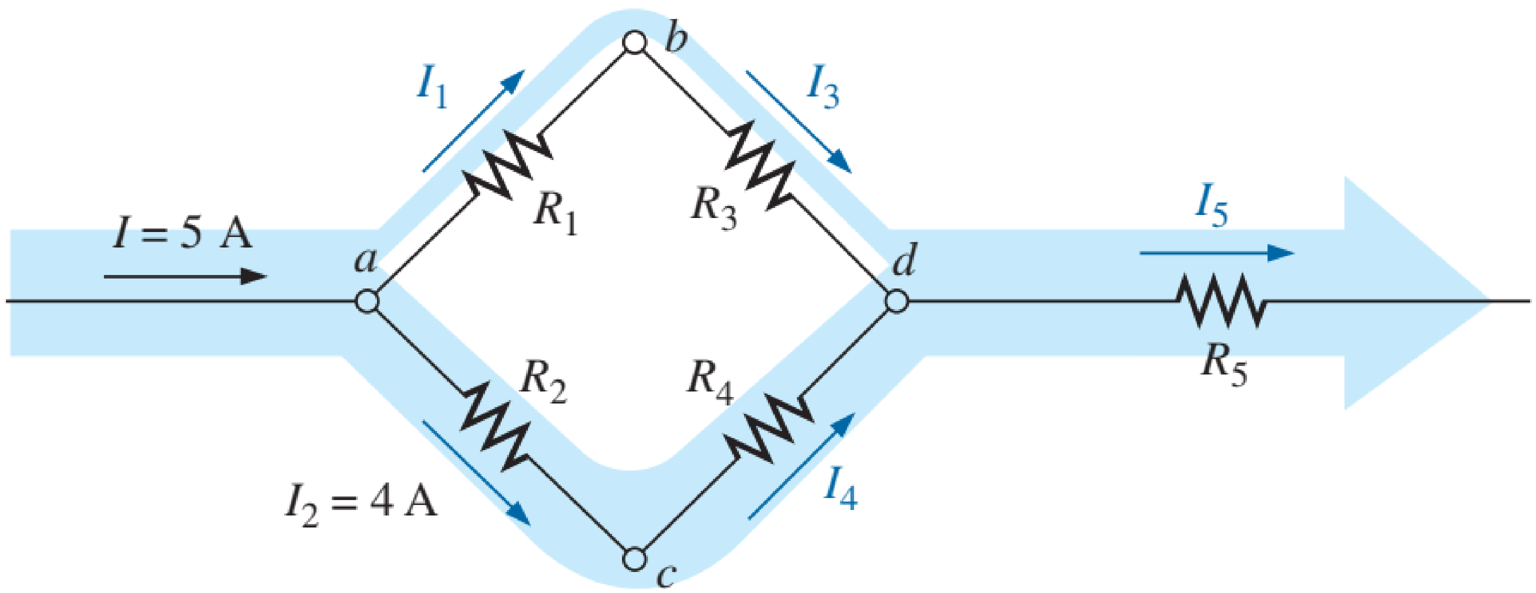



***Figure 12: Four-node configuration for Example 4.***

# References: Textbooks


1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12$^{th}$ Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5$^{th}$ Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6$^{th}$ Edition. McGraw Hill Education.*